

BI-SCUDO Regression - A novel algorithm to infer clinical variables from omics datasets

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In omics data analysis, traditional methods primarily address classification tasks, often leading to a loss of detail during patient stratification due to the discretization of continuous clinical variables.

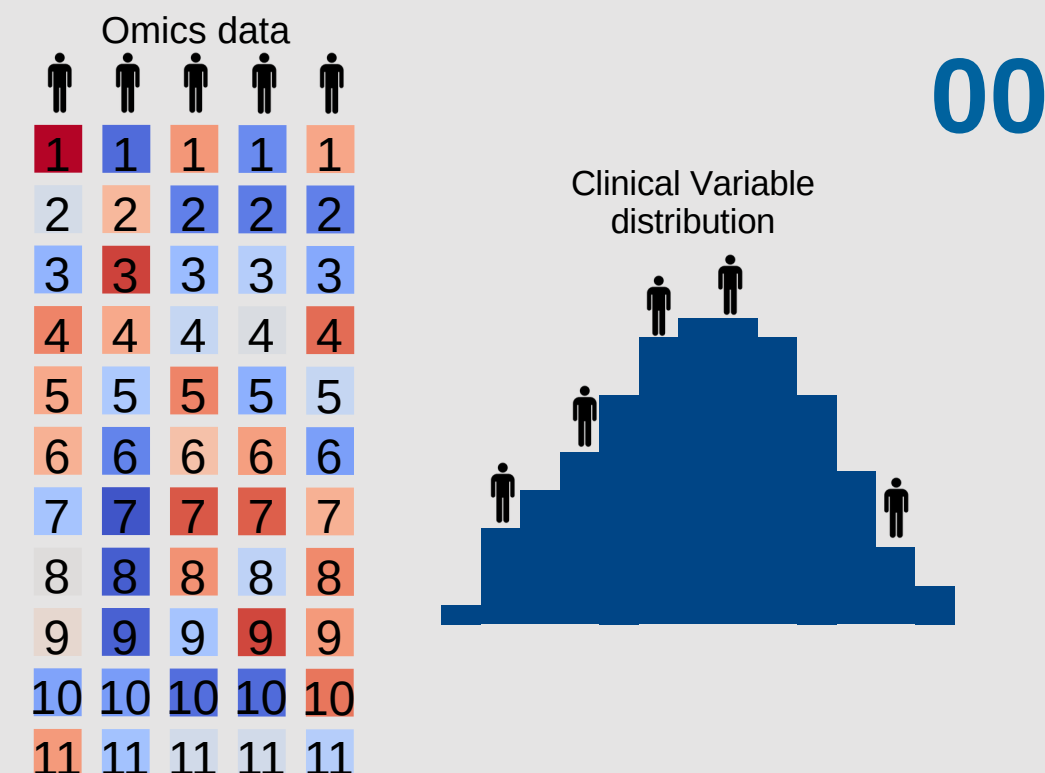
While regression techniques like LASSO and Random Forest Regression exist, they struggle with the noisy nature of omics datasets. In contrast, rank-based methods have shown to be able to deal with them.

This study introduces BI-SCUDO regression, a novel rank-based approach designed to predict a continuous variable and simultaneously perform biomarker identification.

Method

Input Data

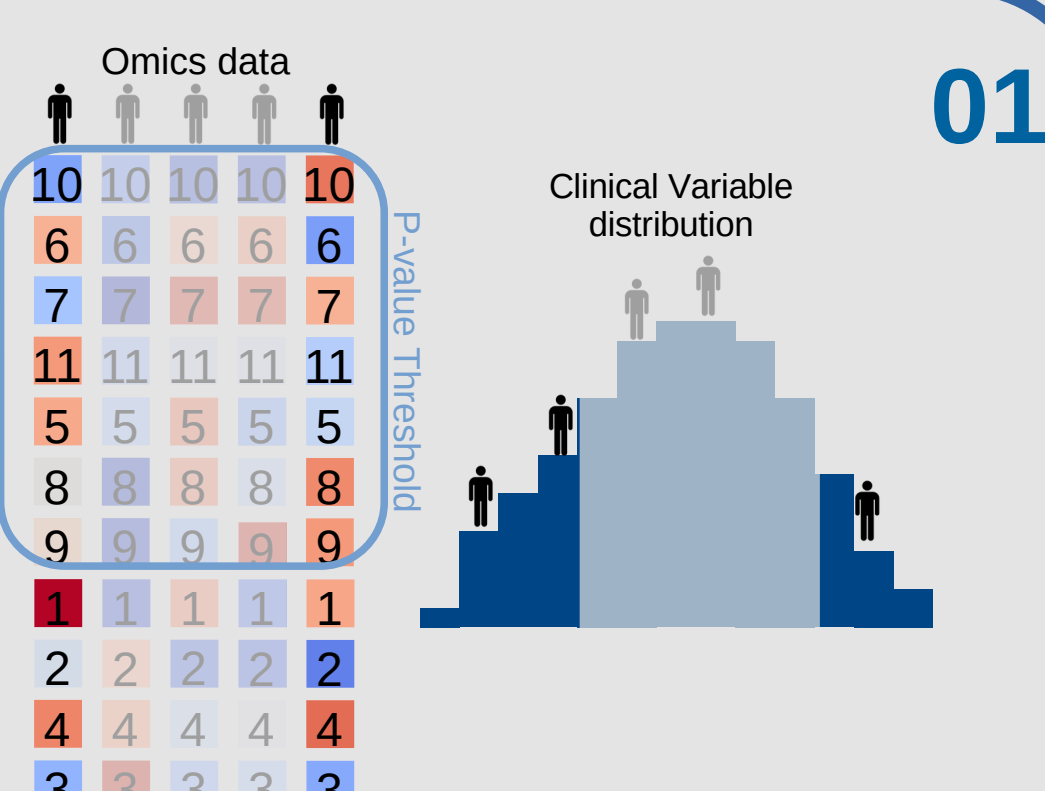
BI-SCUDO regression builds upon a rank-based classification method which demonstrated to be able to handle a variety of omics data types.



00

Statistical Feature Selection

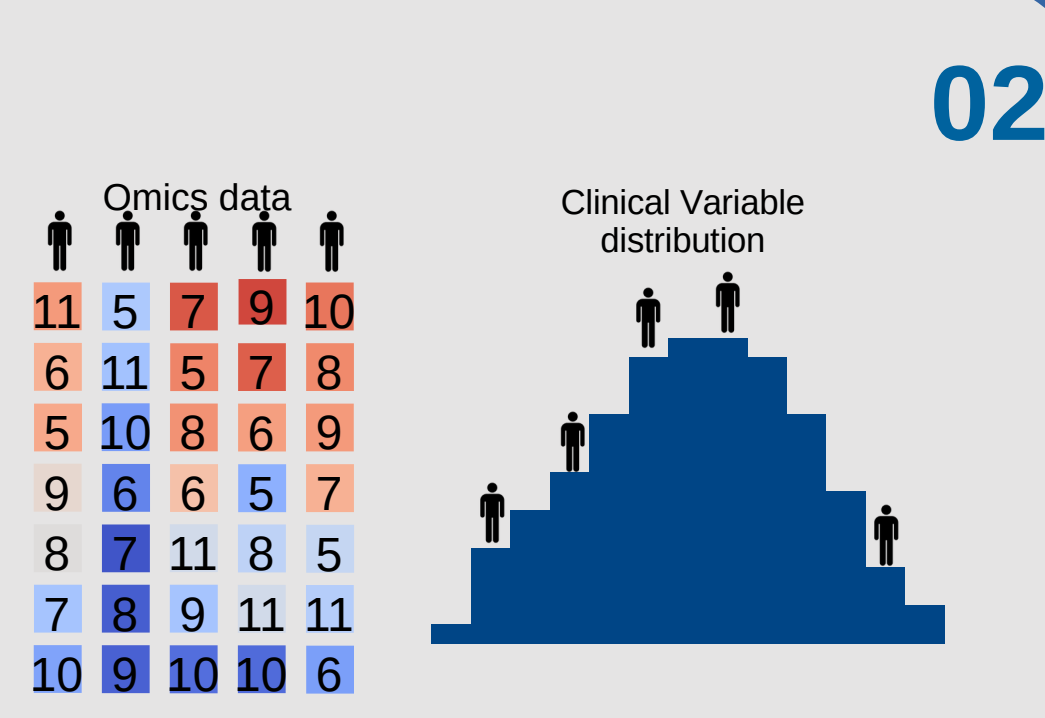
To decrease the dimensionality of the dataset, features are filtered using a Wilcoxon test between classes defined by the extreme values of the clinical variable.



01

Gene Ranking

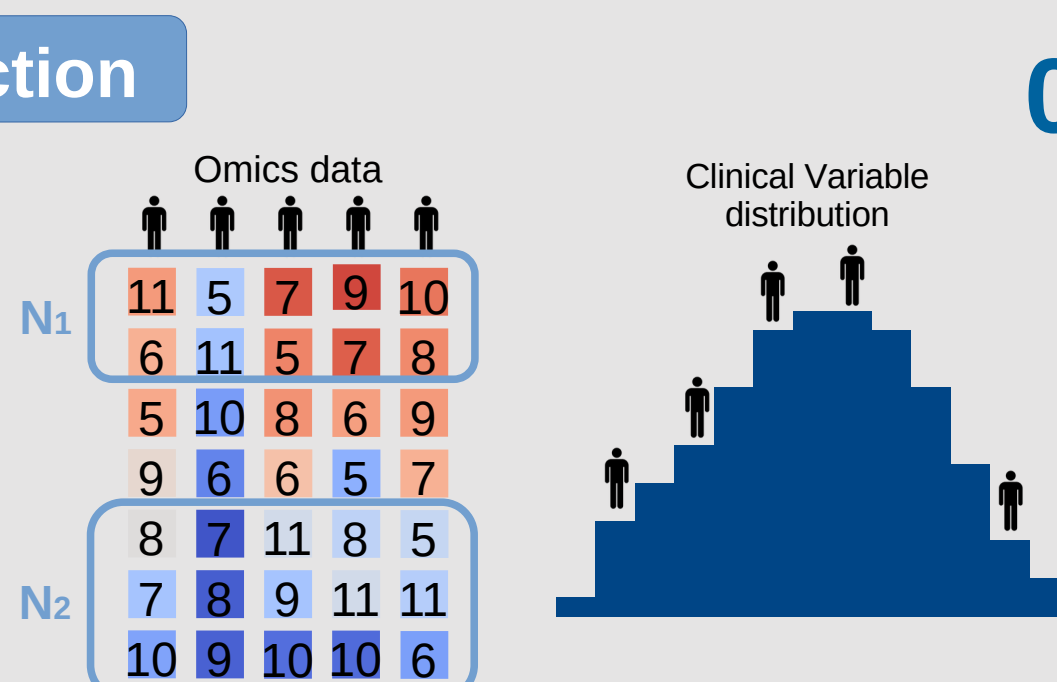
Gene expression value is substituted with its per-sample rank. This makes the method both normalization free and more resilient to noise.



02

Signature extraction

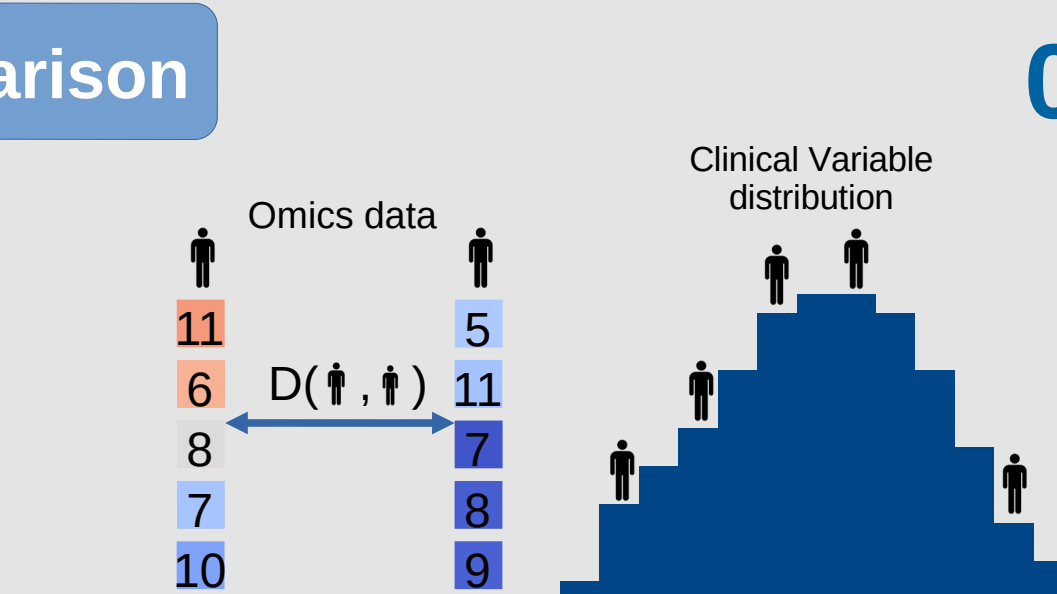
Only the top and bottom-ranked features are further considered, as they represent the most over or under-expressed features.



03

Signature Comparison

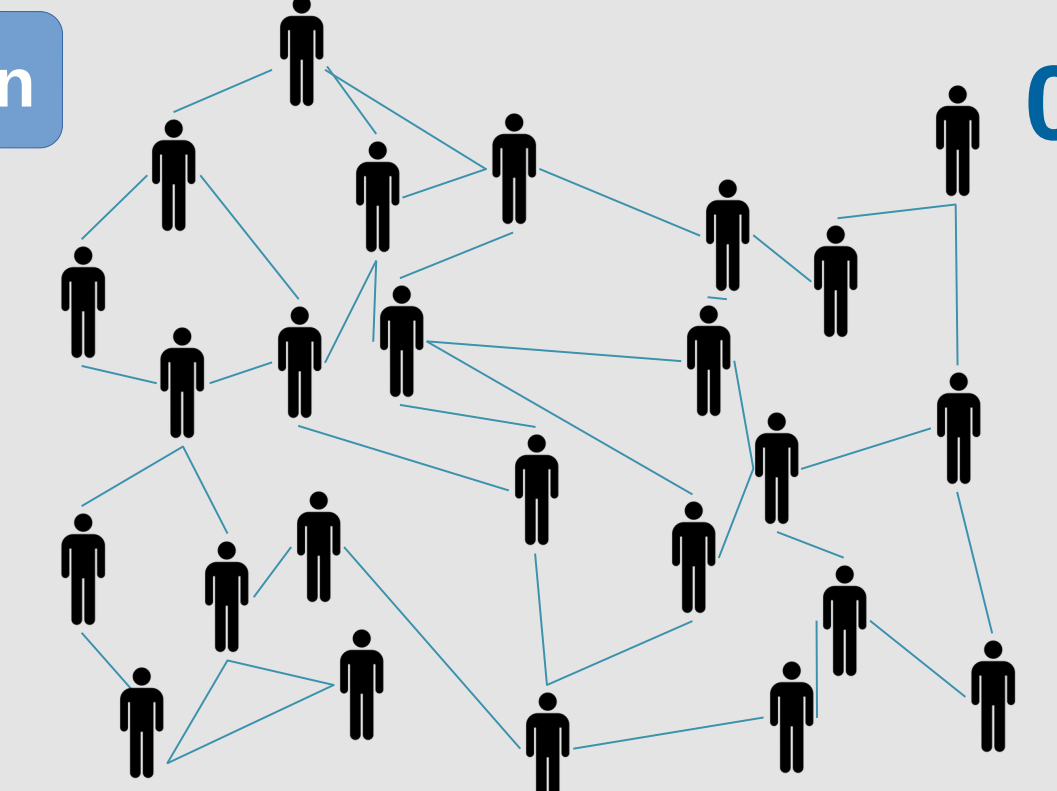
The patient-specific signatures are then compared via an *ad hoc* distance measure based on the enrichment score.



04

Map Construction

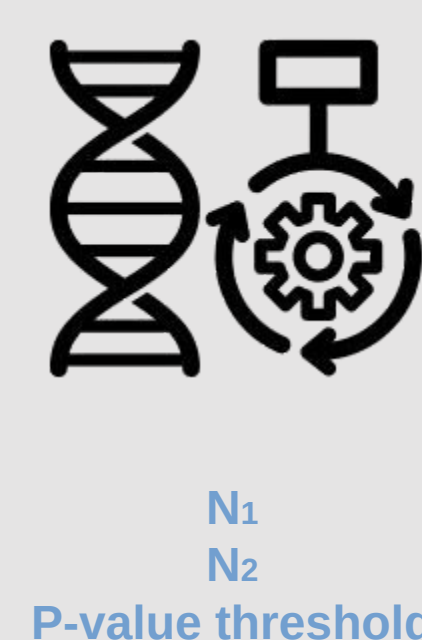
A similarity graph is built from the distances between signatures. This graph will be used to identify sample neighbors in the next steps.



05

Optimization

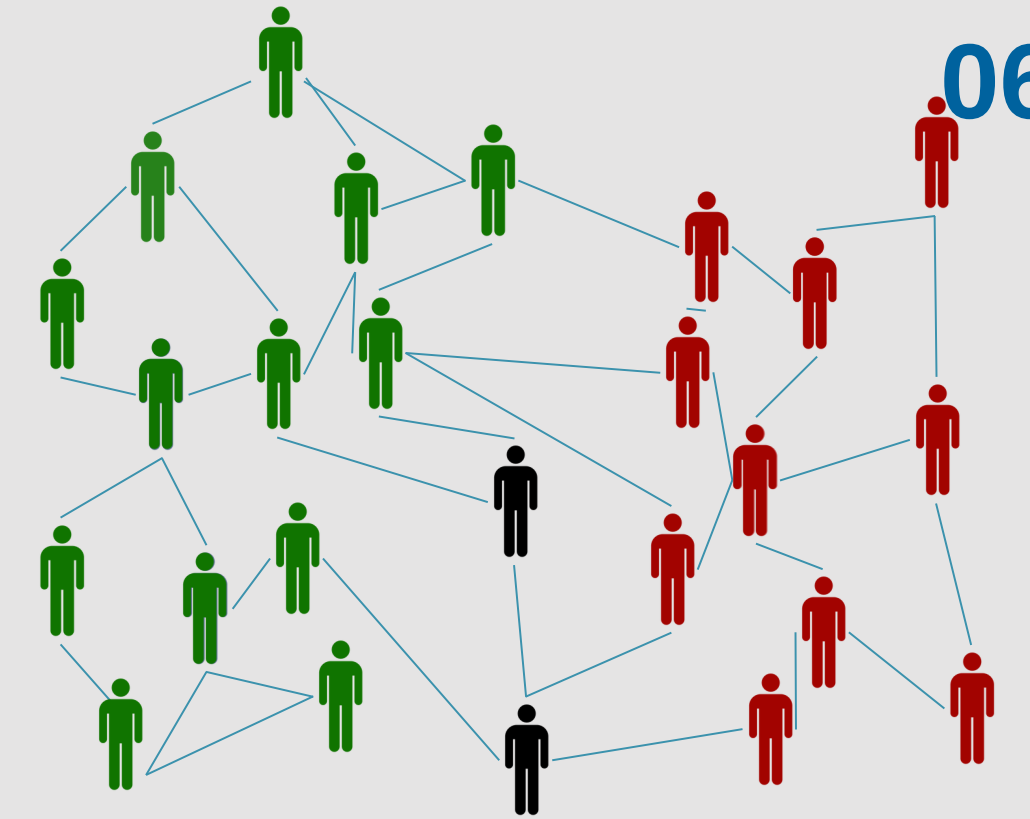
A genetic algorithm is used to identify an optimal biomarker by optimizing the length of the signature and the threshold of the statistical feature selection.



N₁
N₂
P-value threshold

Classification

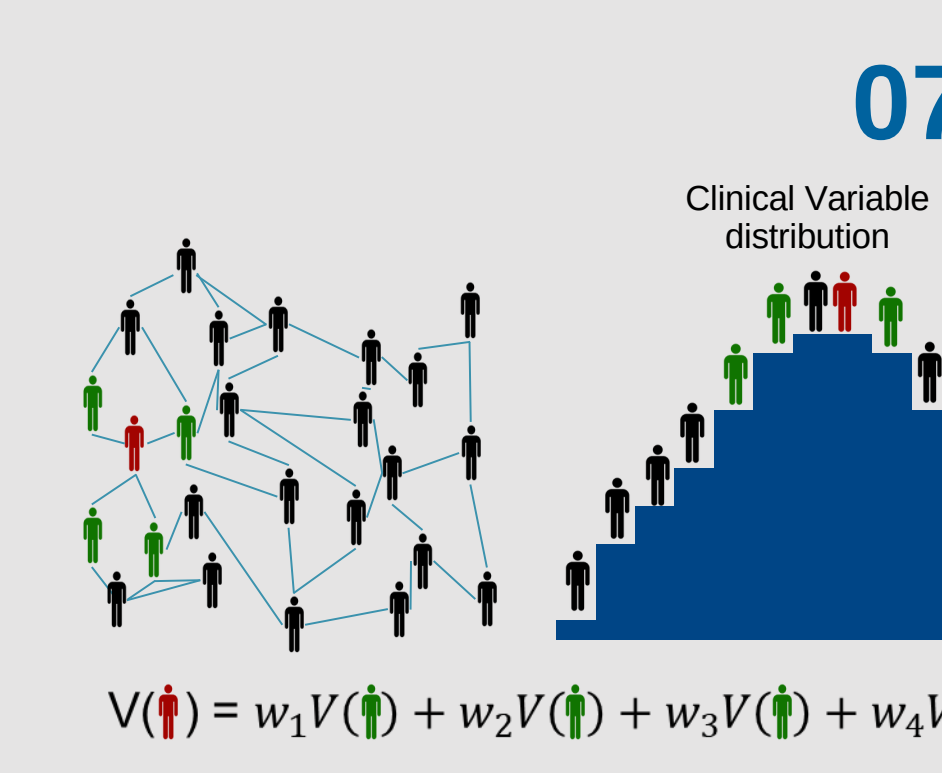
If needed, a patient classification can be performed by identifying clusters in the similarity graph. The position of a new sample in the graph will determine its class.



06

Regression

The value of the clinical variable of a new sample is predicted as the average of the values of its neighbors in the graph, weighted by the distance.



07

Synthetic Data

The method has been tested on a range of *in silico* datasets generated to replicate real scenarios of decreasing signal to noise ratio.

Data generation

Omics features are generated according to the distribution of microarray data and to correlate, or non-correlate with the metadata.

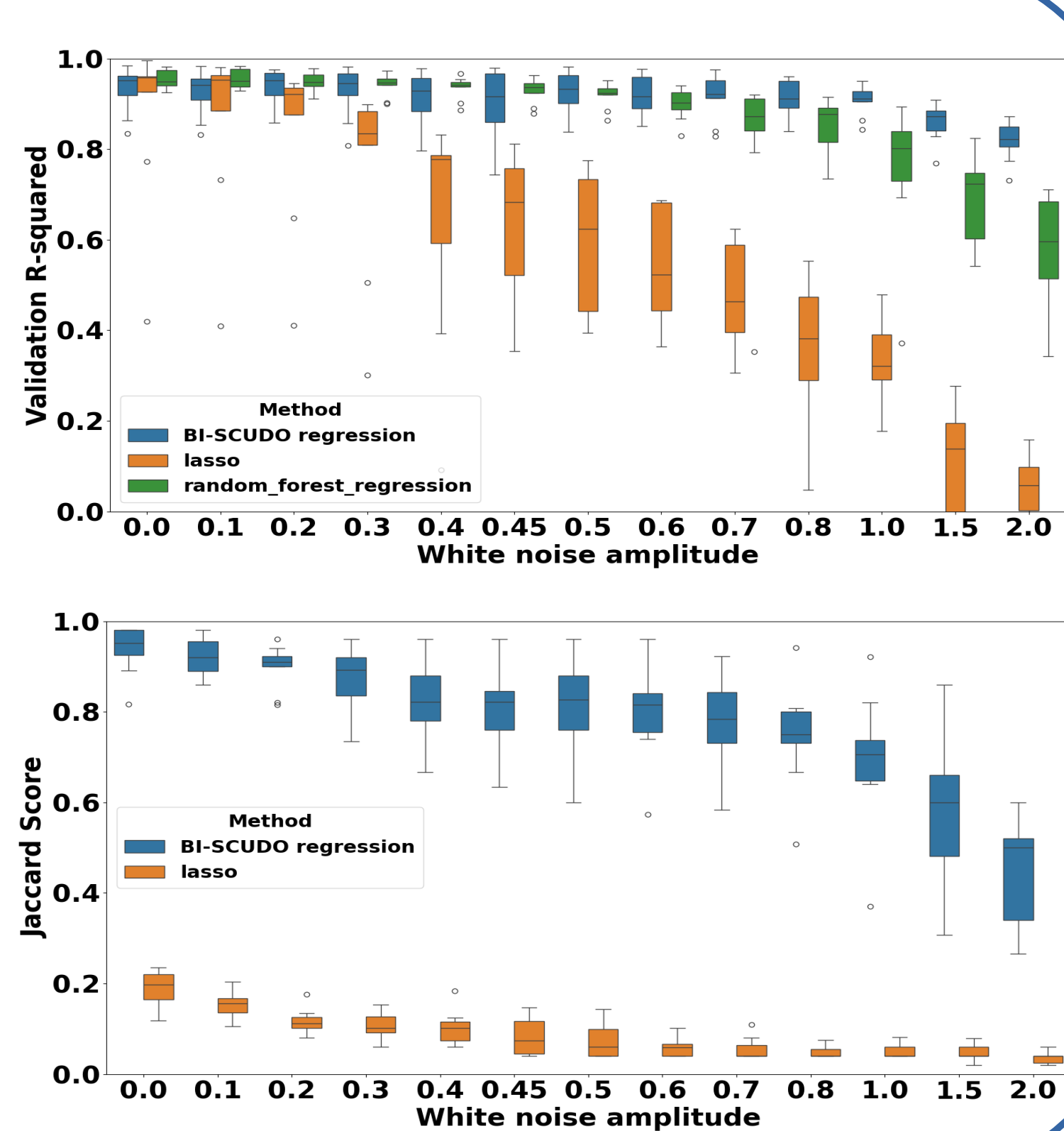
White noise is added to the dataset with increasing amplitude.

BI-SCUDO regression performance was then tested against 10 independent datasets.

Results

BI-SCUDO was able to deal with noise better than both Random Forest Regression And LASSO.

LASSO results were overly restrictive in selecting relevant information, while BI-SCUDO regression was able to capture more relevant information.

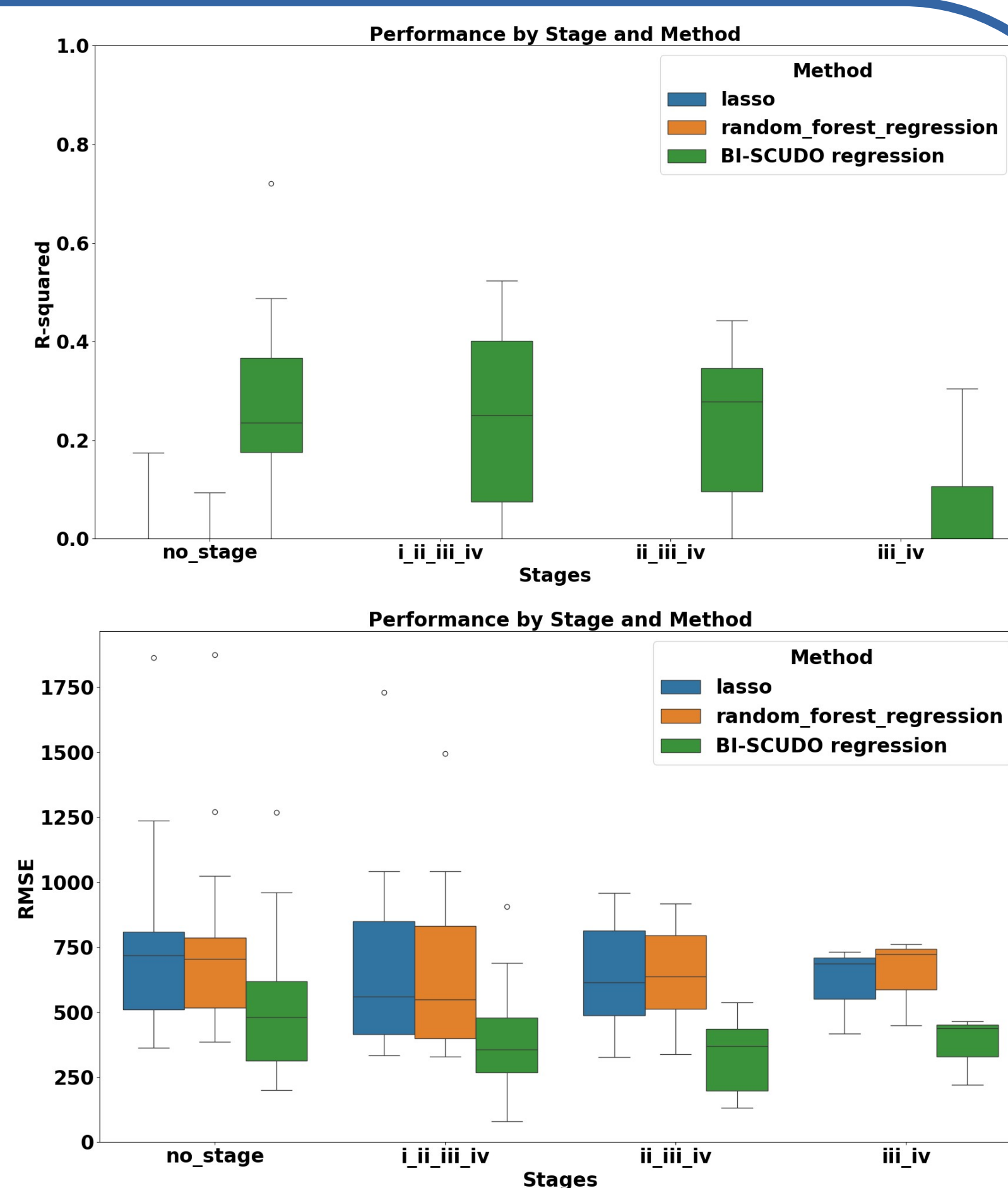


TCGA Data

A real-world assessment has been performed by predicting patients' overall survival from transcriptomic data from public TCGA (The Cancer Genome Atlas) projects [4], with at least 30 samples (32 projects, 3464 samples) stratified via the overall AJCC pathologic stage.

BI-SCUDO regression was again tested against LASSO and Random Forest Regression, evaluating R-squared and root mean square error (RMSE) following from a 10-fold cross validation schema.

BI-SCUDO regression resulted to be the only method able to obtain a positive R-squared in the different datasets. Similarly, considering RSME, BI-SCUDO regression outperformed both LASSO and Random Forest Regression, which exhibited a similar error distribution across the different datasets.



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Conclusions

This study introduces BI-SCUDO regression as a promising alternative to existing methods. The method has shown to be able to outperform state of the art algorithms like LASSO and Random Forest Regression on synthetic and real experimental data.

- ✓ BI-SCUDO Regression is able to predict with high accuracy the objective variable.
- ✓ BI-SCUDO Regression is able to extract meaningful information from the dataset, allowing to infer the biological processes underlying the distribution of the objective variable.

Next Steps

- Further validate the performance of our method on additional independent transcriptomic datasets.
- Assess the applicability of our method on other kinds of omics data, such as single cell and metagenomic datasets.
- The algorithm is currently undergoing runtime optimization and preparation for deployment.

References

- [1] Li, G., Zrimec, J., Ji, B., Geng, J., Larsbrink, J., Zelezniak, A., Nielsen, J., & Engqvist, M. K. (2021). Performance of Regression Models as a Function of Experiment Noise. In *Bioinformatics and Biology Insights* (Vol. 15). SAGE Publications. <https://doi.org/10.1177/11779322211020315>
- [2] Jerley, G., & Kostinski, A. (2019). Universal Rank-Order Transform to Extract Signals from Noisy Data. In *Physical Review X* (Vol. 9, Issue 3). American Physical Society (APS). <https://doi.org/10.1103/physrevx.9.031039>
- [3] Lauria, M. (2013). Rank-based transcriptional signatures. In *Systems Biomedicine* (Vol. 1, Issue 4, pp. 228–239). Informa UK Limited. <https://doi.org/10.4161/sysb.25982>
- [4] Weinstein, J. N., Collisson, E. A., Mills, G. B., Shaw, K. R. M., Ozenberger, B. A., Ellrott, K., Shmulevich, I., Sander, C., & Stuart, J. M. (2013). The Cancer Genome Atlas Pan-Cancer analysis project. *Nature Genetics*, 45(10), 1113–1120. <https://doi.org/10.1038/ng.2764>